

Lecture 13:

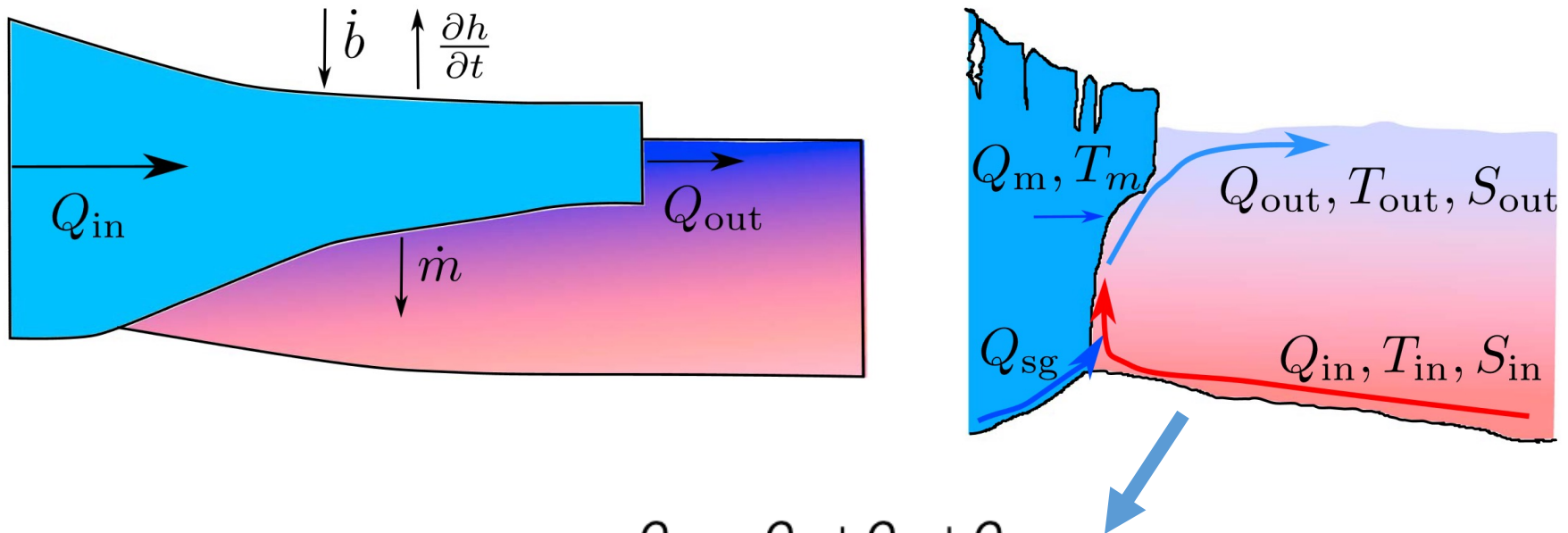
Ice-ocean interaction

- Roles of the ocean in mass balance
 - direct removal by melting
 - at calving fronts and beneath ice shelves
 - plumes
 - Allowing iceberg calving
 - Thinning ice shelves reducing buttressing
- Other
 - Tidal oscillations in ice speed
 - Tidal pumping
 - An update on Flask Glacier

Where glaciers meet water: Subaqueous melt and its relevance to glaciers in various settings

Martin Truffer¹ and Roman J. Motyka¹

Two methods for estimating subaqueous melting

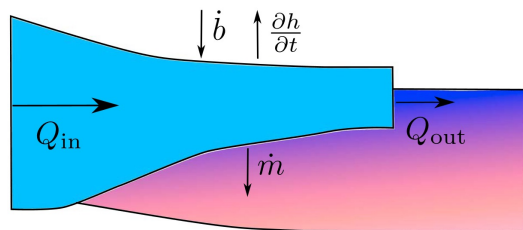
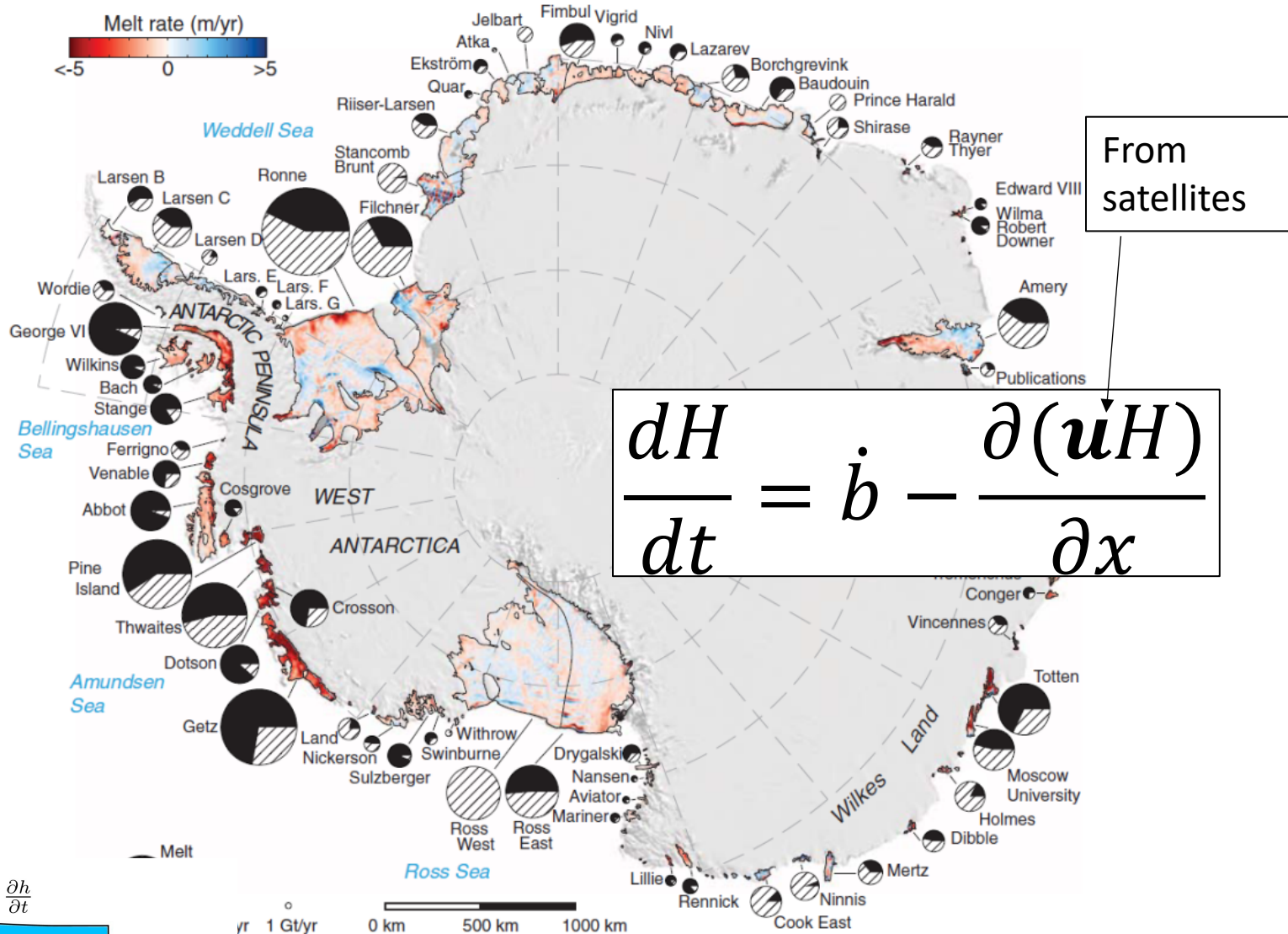


$$Q_{out} = Q_{in} + Q_{sg} + Q_m,$$

$$Q_{out}C_w(T_{out} - T_{ref}) = Q_{in}C_w(T_{out} - T_{ref}) + Q_m(T_m - T_{ref}) - Q_m(L_f + C_i(T_m - T_i)),$$

$$Q_{out}S_{out} = Q_{in}S_{in}.$$

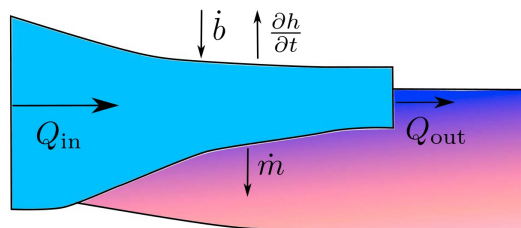
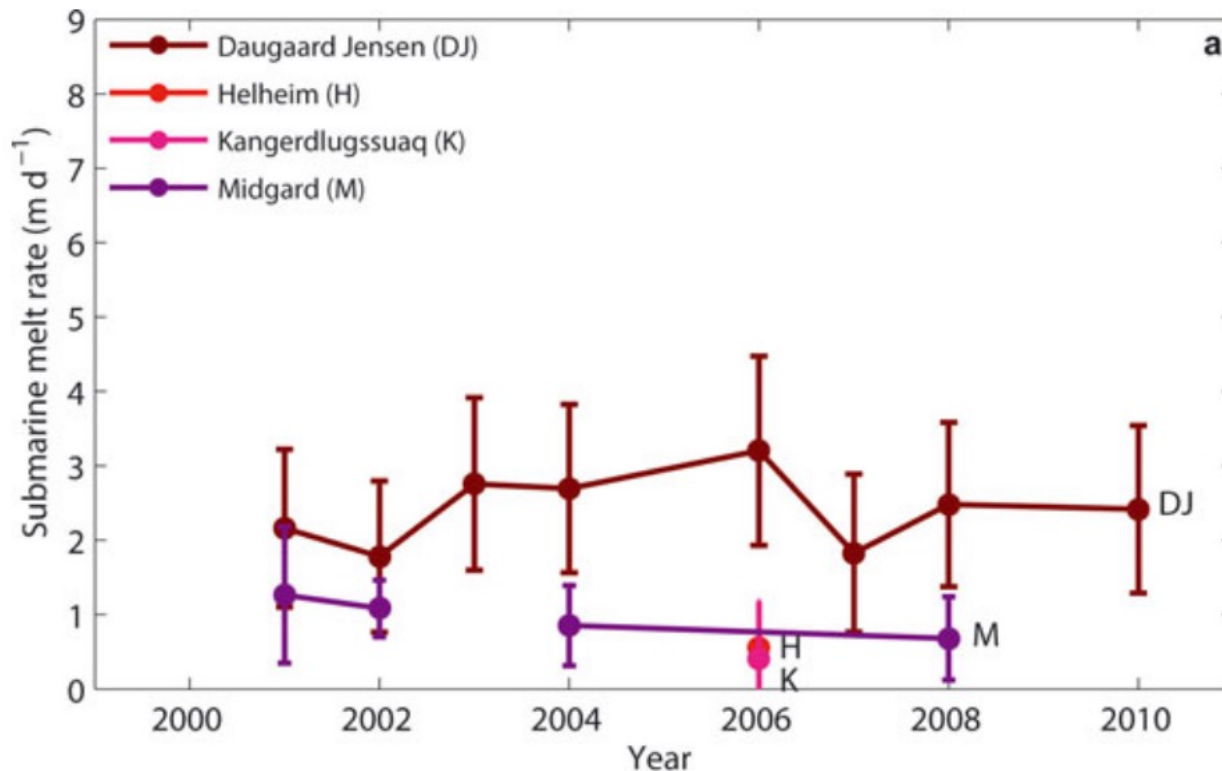
Ice Shelf mass balance



Rignot et al. (2013)

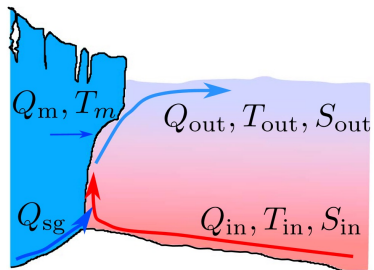
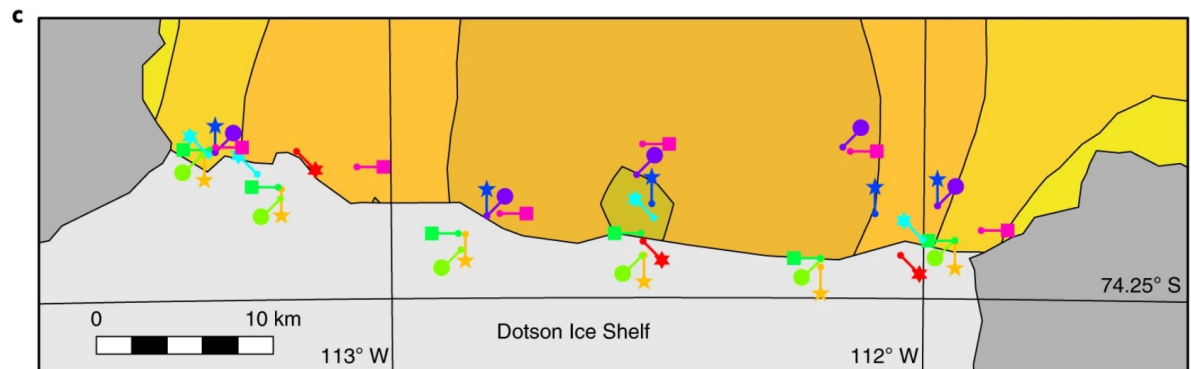
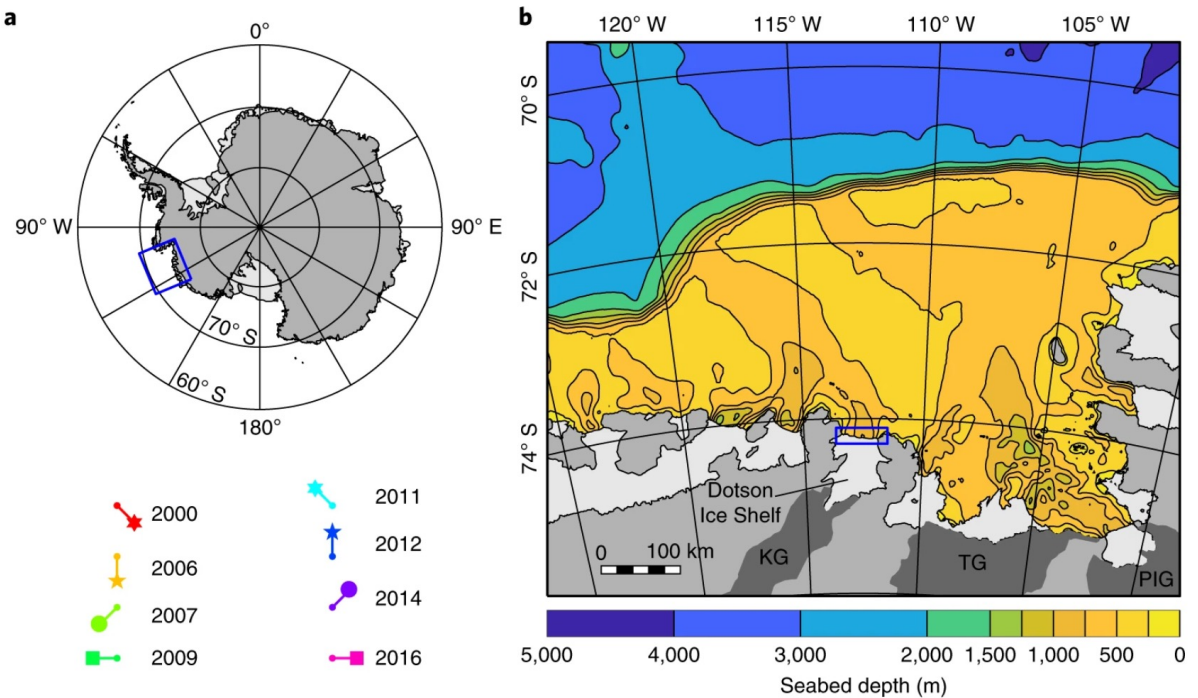
Submarine melt rate estimates for floating termini of Greenland outlet glaciers (2000–2010)

Ellyn M. ENDERLIN, Ian M. HOWAT



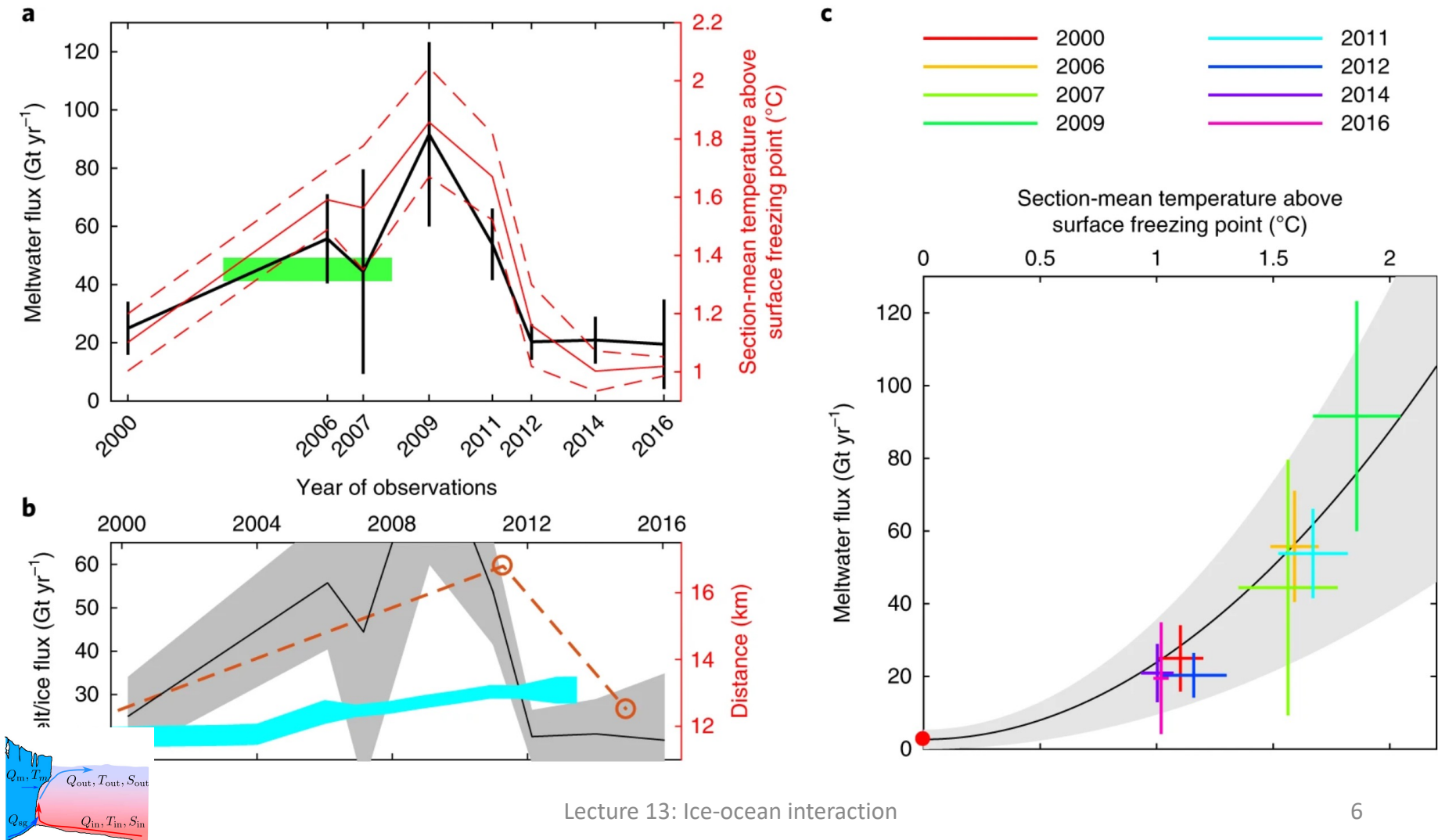
West Antarctic Ice Sheet retreat in the Amundsen Sea driven by decadal oceanic variability

Adrian Jenkins^{1*}, Deb Shoosmith¹, Pierre Dutrieux², Stan Jacobs², Tae Wan Kim³, Sang Hoon Lee³,
Ho Kyung Ha⁴ and Sharon Stammerjohn⁵



West Antarctic Ice Sheet retreat in the Amundsen Sea driven by decadal oceanic variability

Adrian Jenkins^{1*}, Deb Shoosmith¹, Pierre Dutrieux², Stan Jacobs², Tae Wan Kim³, Sang Hoon Lee³, Ho Kyung Ha⁴ and Sharon Stammerjohn⁵



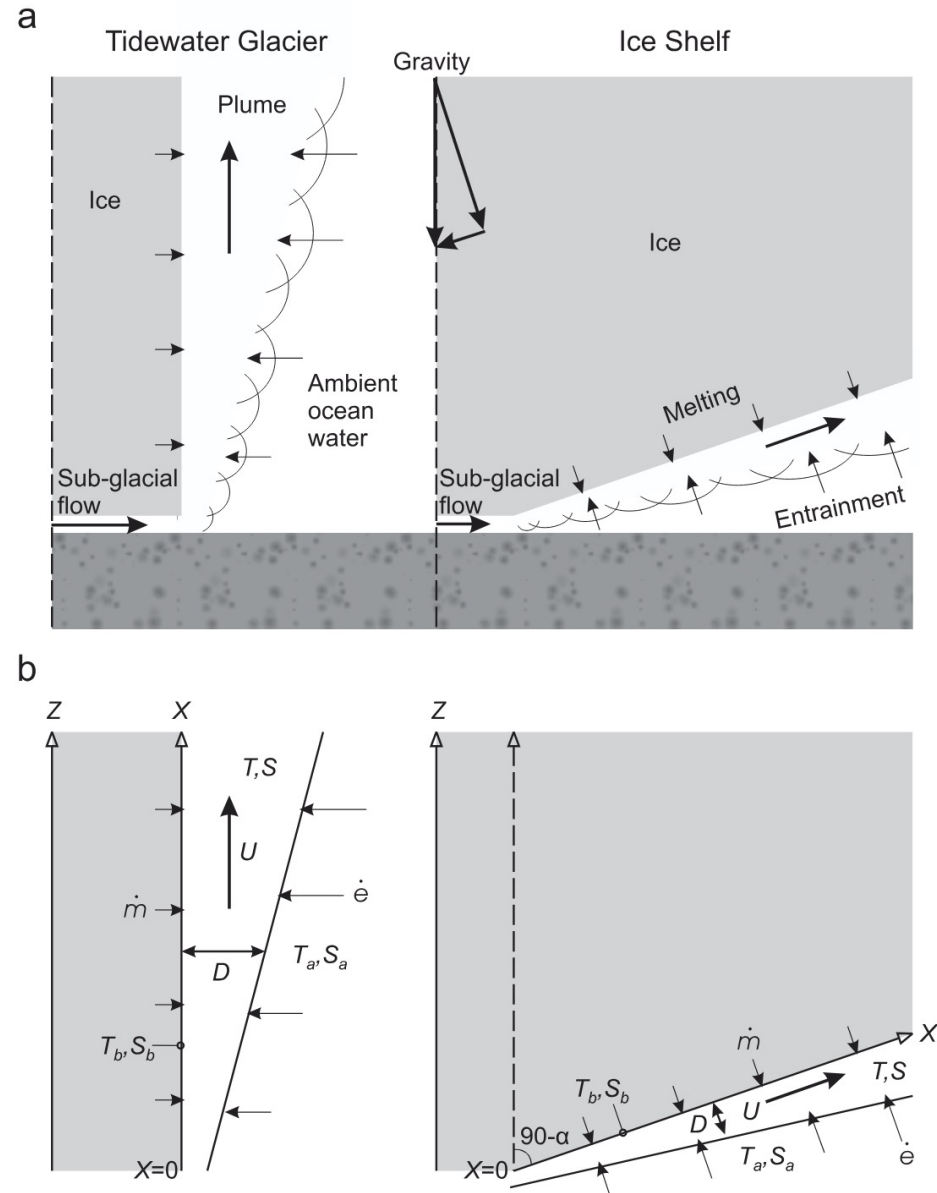
Plume theory

Convection-Driven Melting near the Grounding Lines of Ice Shelves and Tidewater Glaciers

ADRIAN JENKINS

British Antarctic Survey, Natural Environment Research Council, Cambridge, United Kingdom

D is plume thickness
 T is temperature
 S is salinity
 U is speed
 X is distance along the plume



Plume theory

Convection-Driven Melting near the Grounding Lines of Ice Shelves and Tidewater Glaciers

ADRIAN JENKINS

British Antarctic Survey, Natural Environment Research Council, Cambridge, United Kingdom

D is plume thickness

T is temperature

S is salinity

U is speed

X is distance along the plume

$$\frac{d}{dX}(DU) = \dot{e} + \dot{m}, \quad (1)$$

$$\frac{d}{dX}(DU^2) = D \left(\frac{\rho_a - \rho}{\rho_0} \right) g \sin \alpha - C_d U^2, \quad (2)$$

$$\frac{d}{dX}(DUT) = \dot{e}T_a + \dot{m}T_b - C_d^{1/2} U \Gamma_T (T - T_b), \quad \text{and} \quad (3)$$

$$\frac{d}{dX}(DUS) = \dot{e}S_a + \dot{m}S_b - C_d^{1/2} U \Gamma_S (S - S_b), \quad (4)$$

Plume theory

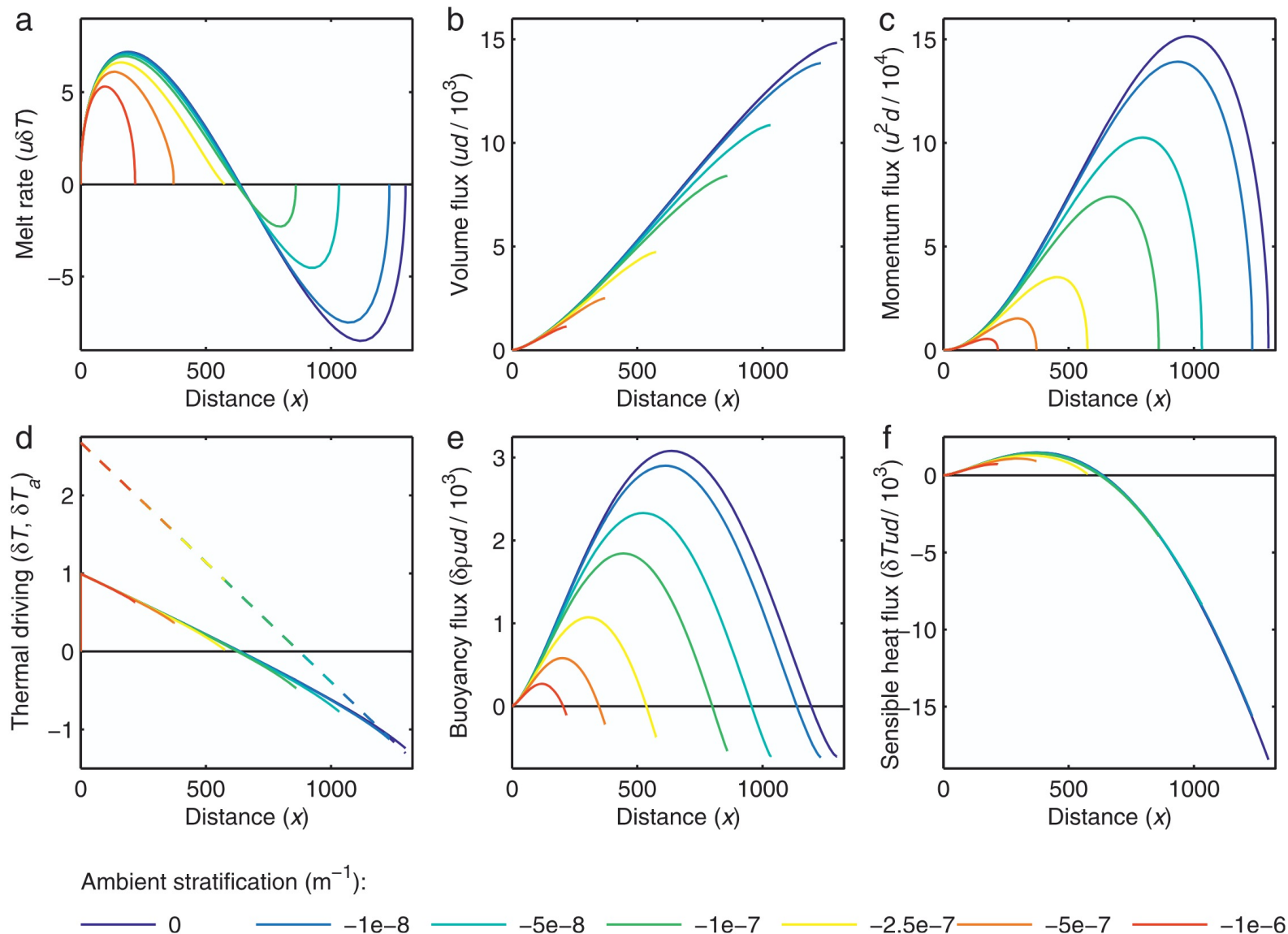
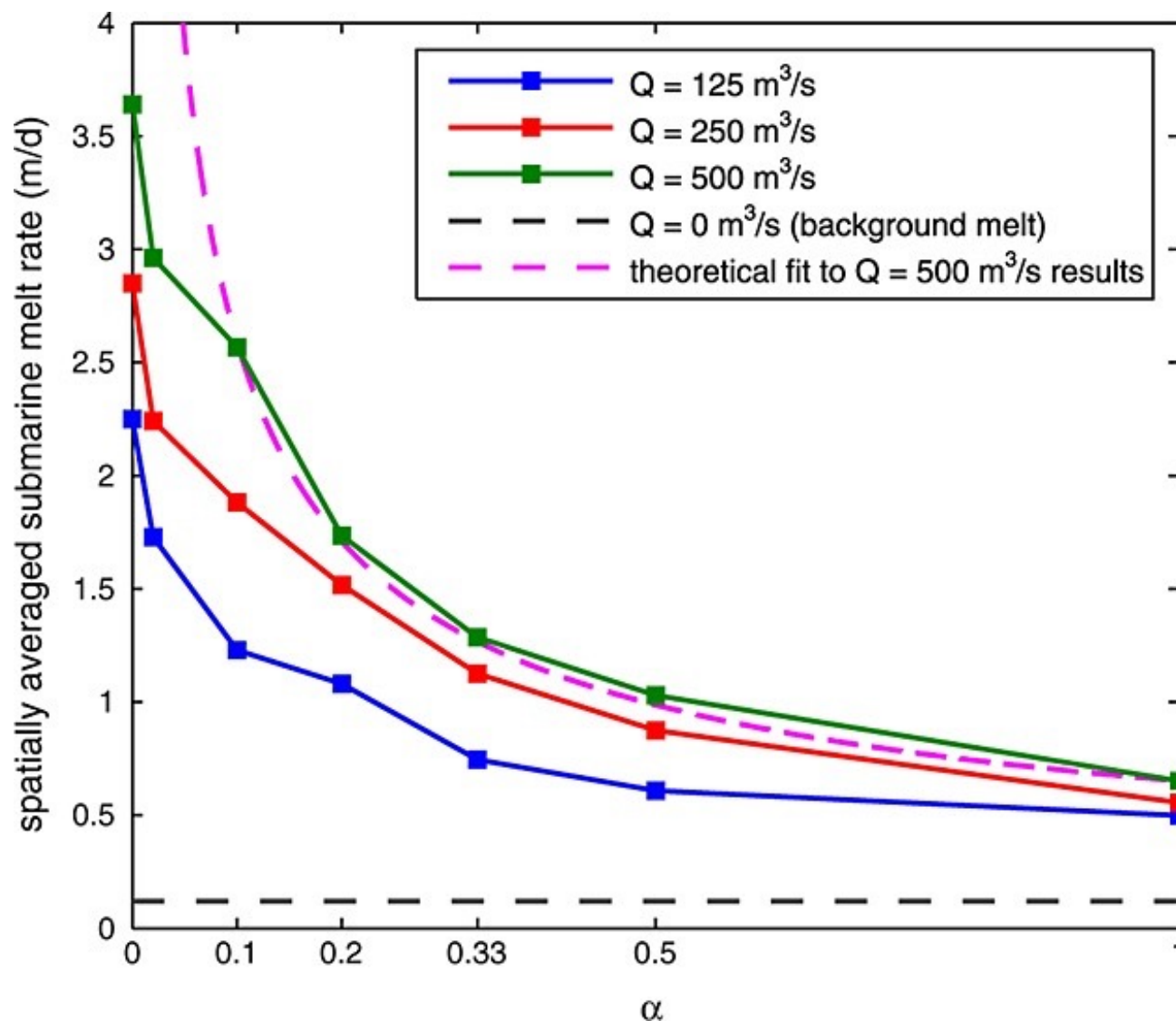


FIG. 2. Dimensionless (a) melt rate, (b) volume flux, (c) momentum flux, (d) thermal driving, (e) buoyancy flux, and (f) sensible heat flux plotted as a function of dimensionless distance for a plume initiated by a freshwater flux of $5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$ beneath an ice shelf with a basal slope ($\sin \alpha$) of 0.01. The ambient ocean has a temperature 2°C above the freezing point at the starting depth, a salinity of 35, and varying stratification indicated by lines of differing color. The resulting ambient thermal driving is indicated by the dashed lines in (d).

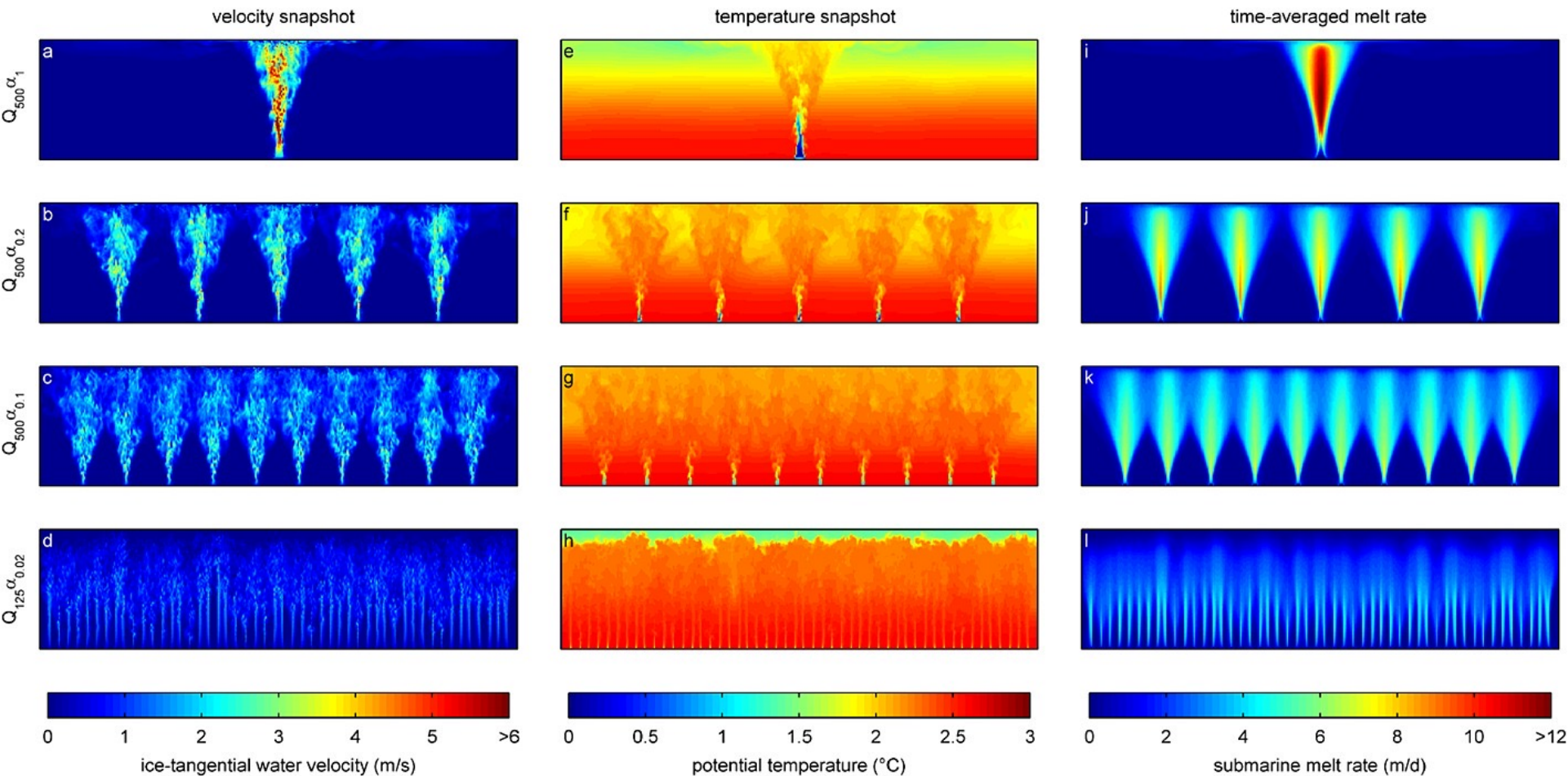
Effect of near-terminus subglacial hydrology on tidewater glacier submarine melt rates

D. A. Slater , P. W. Nienow, T. R. Cowton, D. N. Goldberg, A. J. Sole



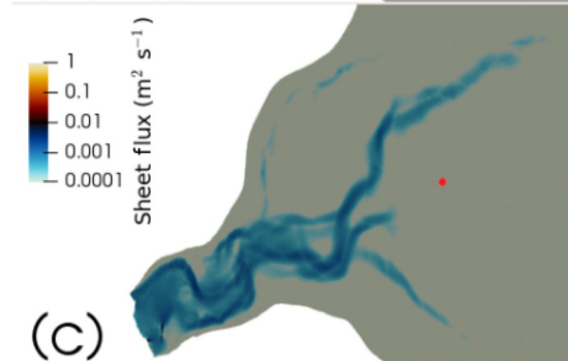
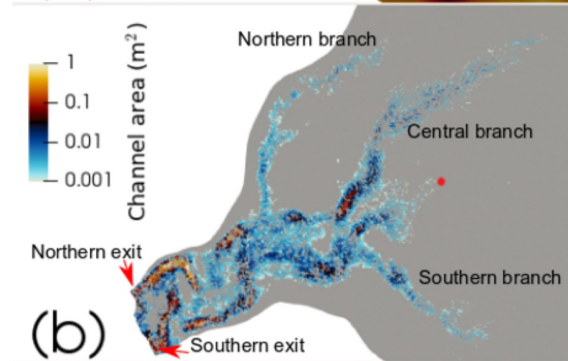
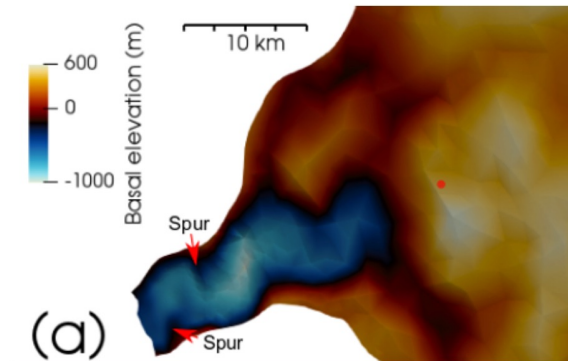
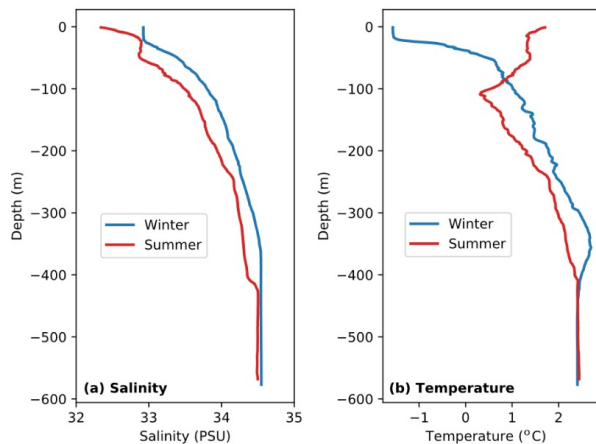
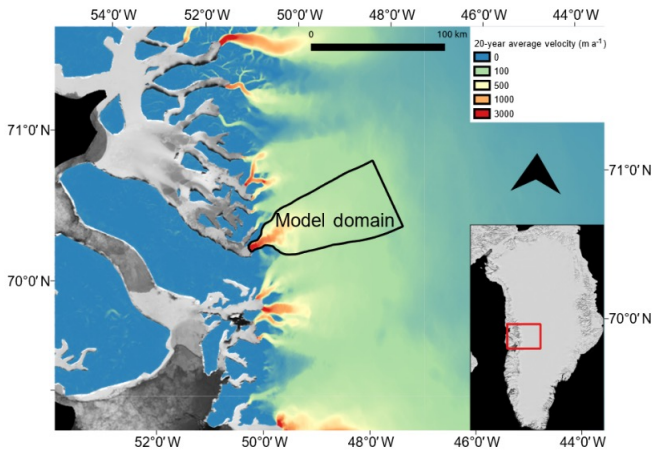
Effect of near-terminus subglacial hydrology on tidewater glacier submarine melt rates

D. A. Slater , P. W. Nienow, T. R. Cowton, D. N. Goldberg, A. J. Sole



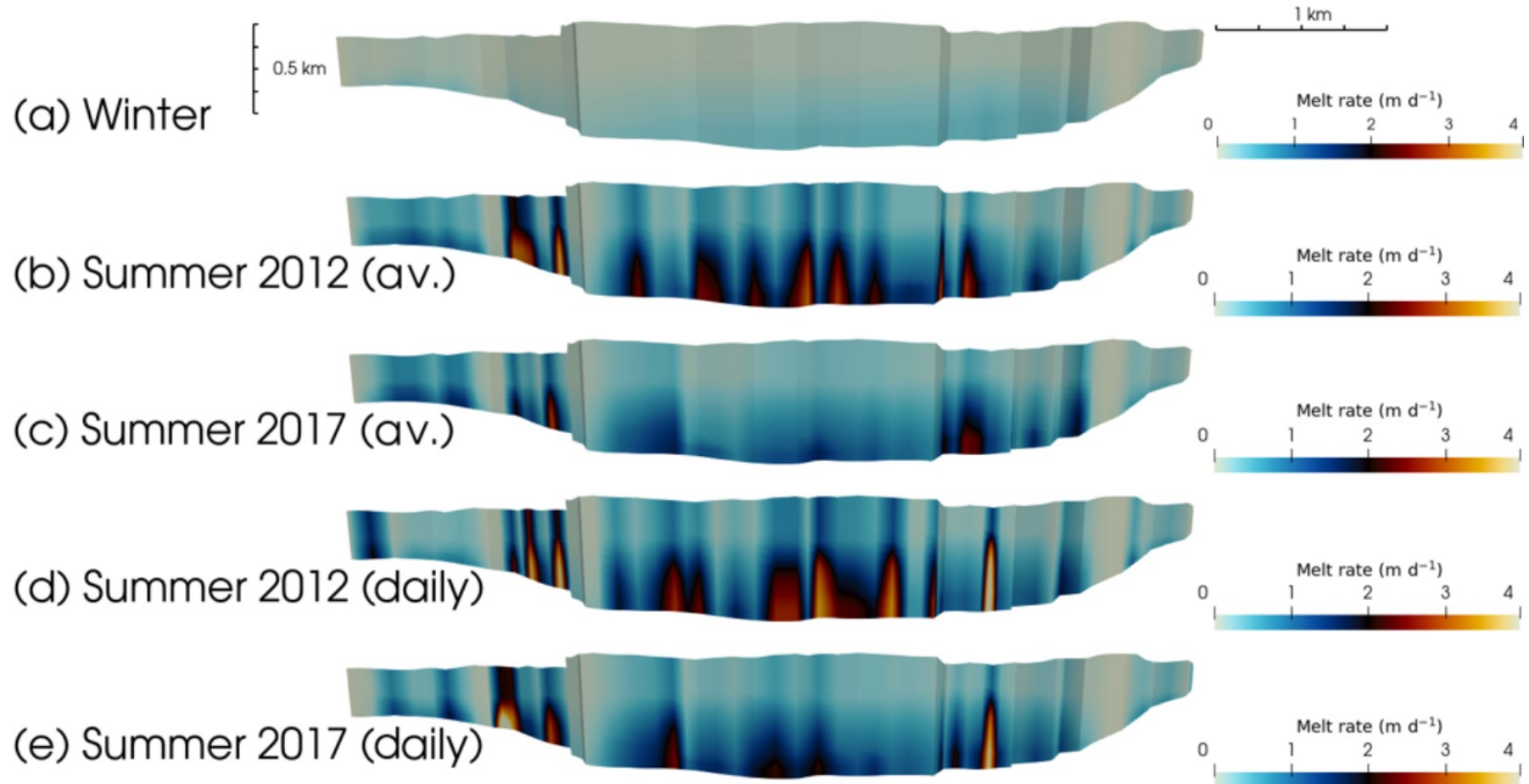
Coupled modelling of subglacial hydrology and calving-front melting at Store Glacier, West Greenland

Samuel J. Cook¹, Poul Christoffersen¹, Joe Todd², Donald Slater³, and Nolwenn Chauché⁴



Coupled modelling of subglacial hydrology and calving-front melting at Store Glacier, West Greenland

Samuel J. Cook¹, Poul Christoffersen¹, Joe Todd², Donald Slater³, and Nolwenn Chauché⁴



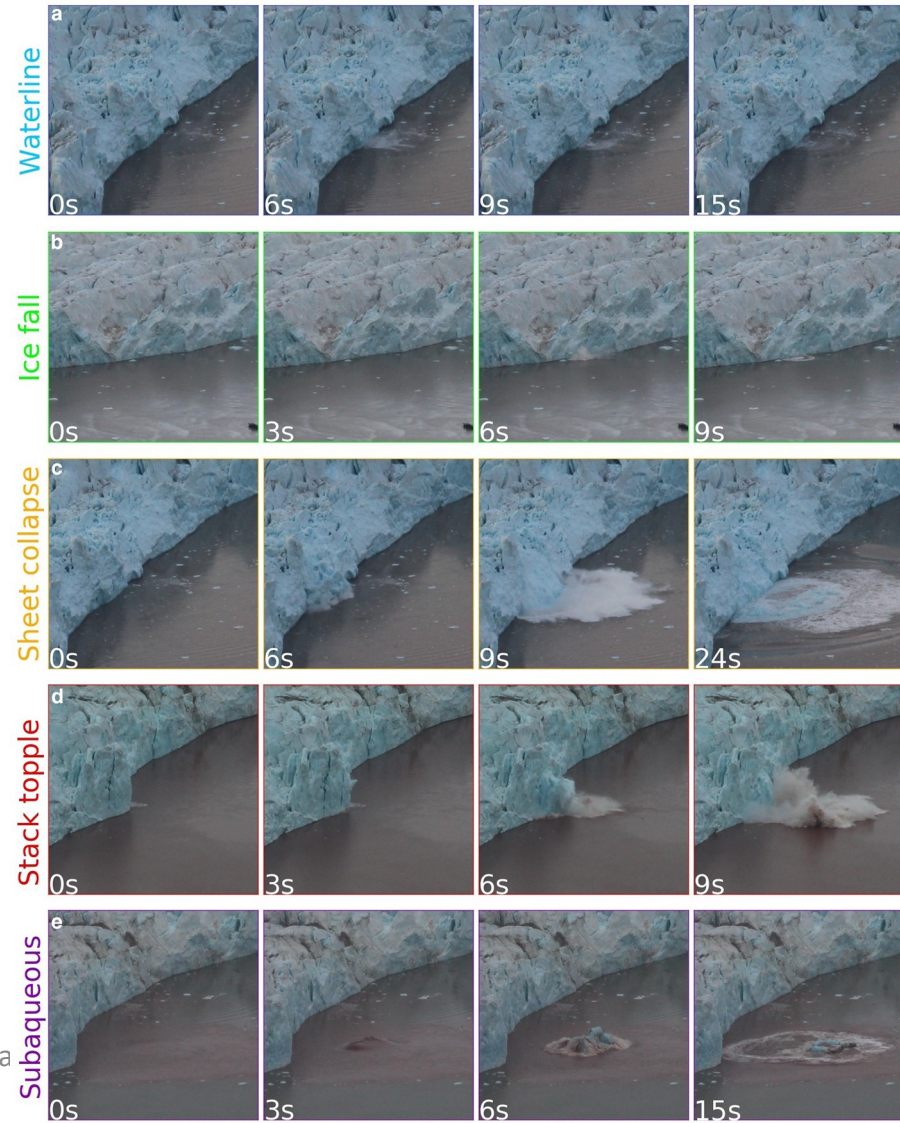
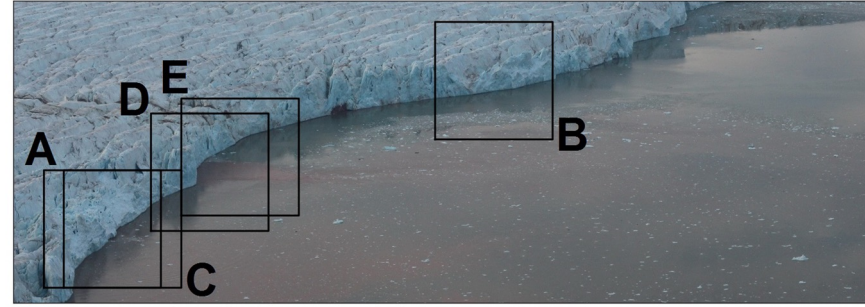
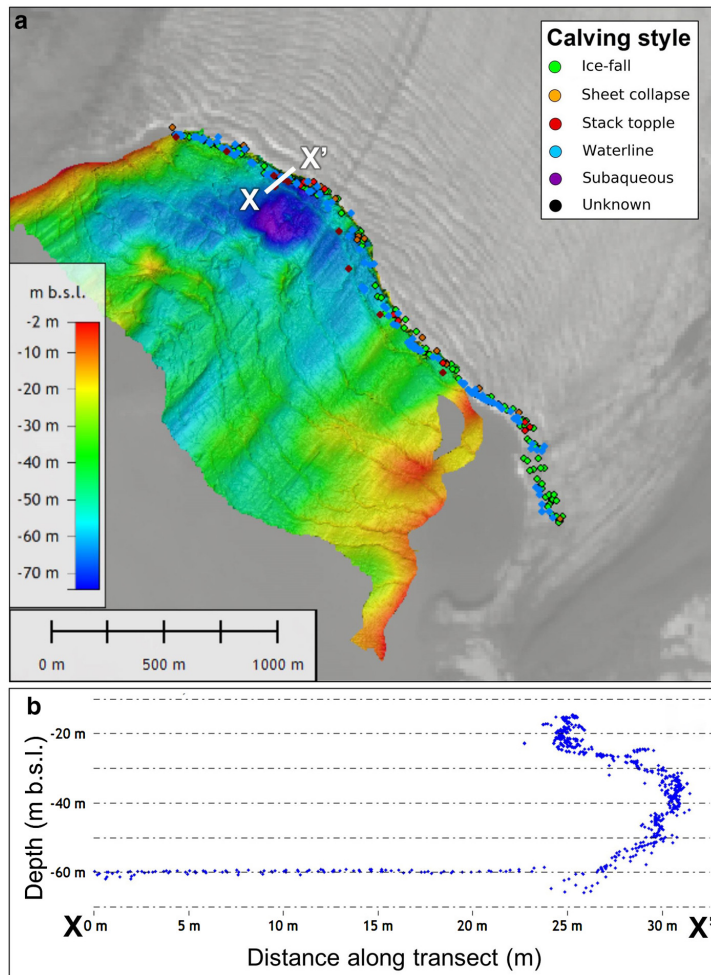
The ocean's role in calving

Calving controlled by melt-under-cutting: detailed calving styles revealed through time-lapse observations

Published online by Cambridge University Press: 30 January 2019

Penelope How , Kristin M. Schild, Douglas I. Benn, Riko Noormets, Nina Kirchner, Adrian Luckman, Dorothee Vallot, Nicholas R. J. Hulton and Chris Borstad

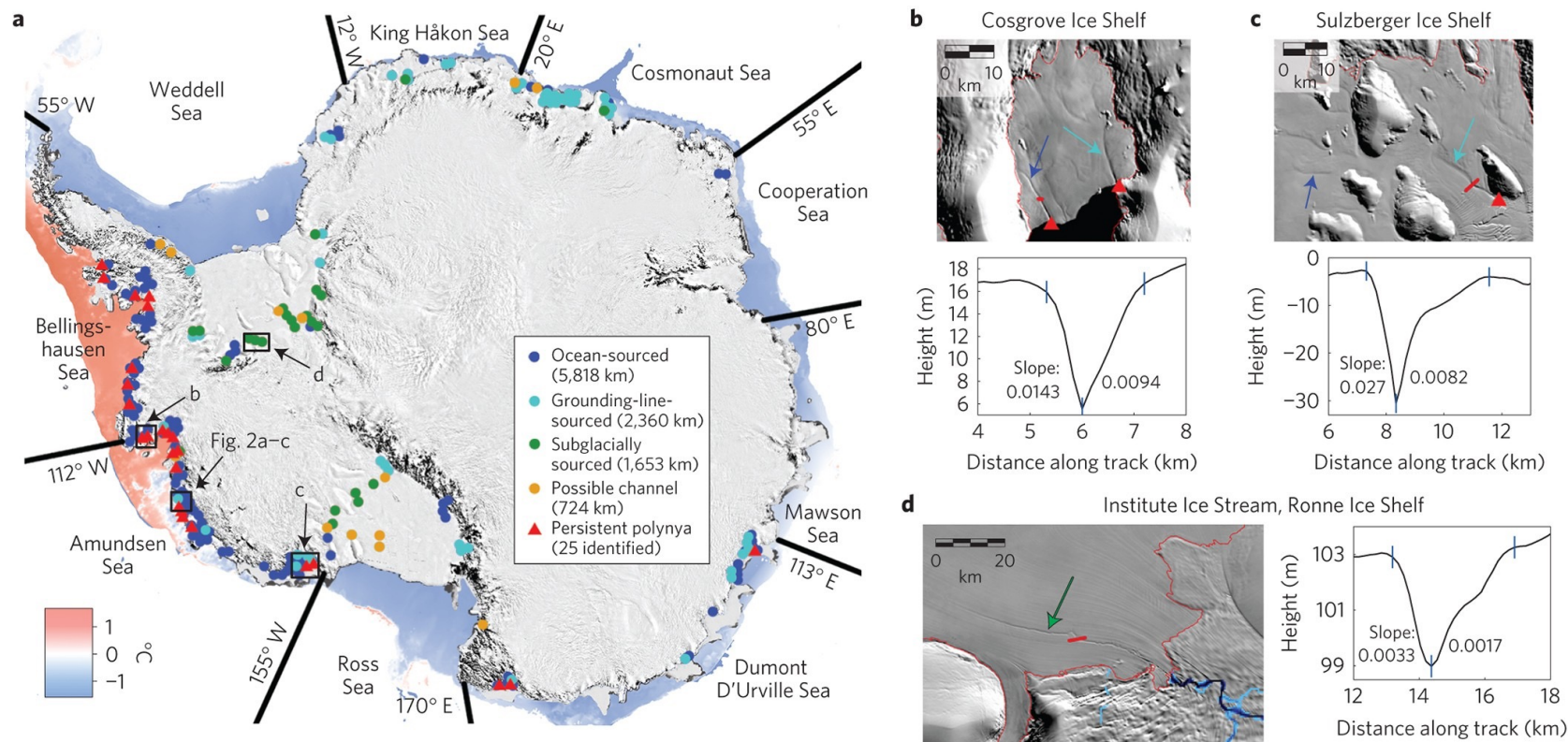
[Show author details](#) 



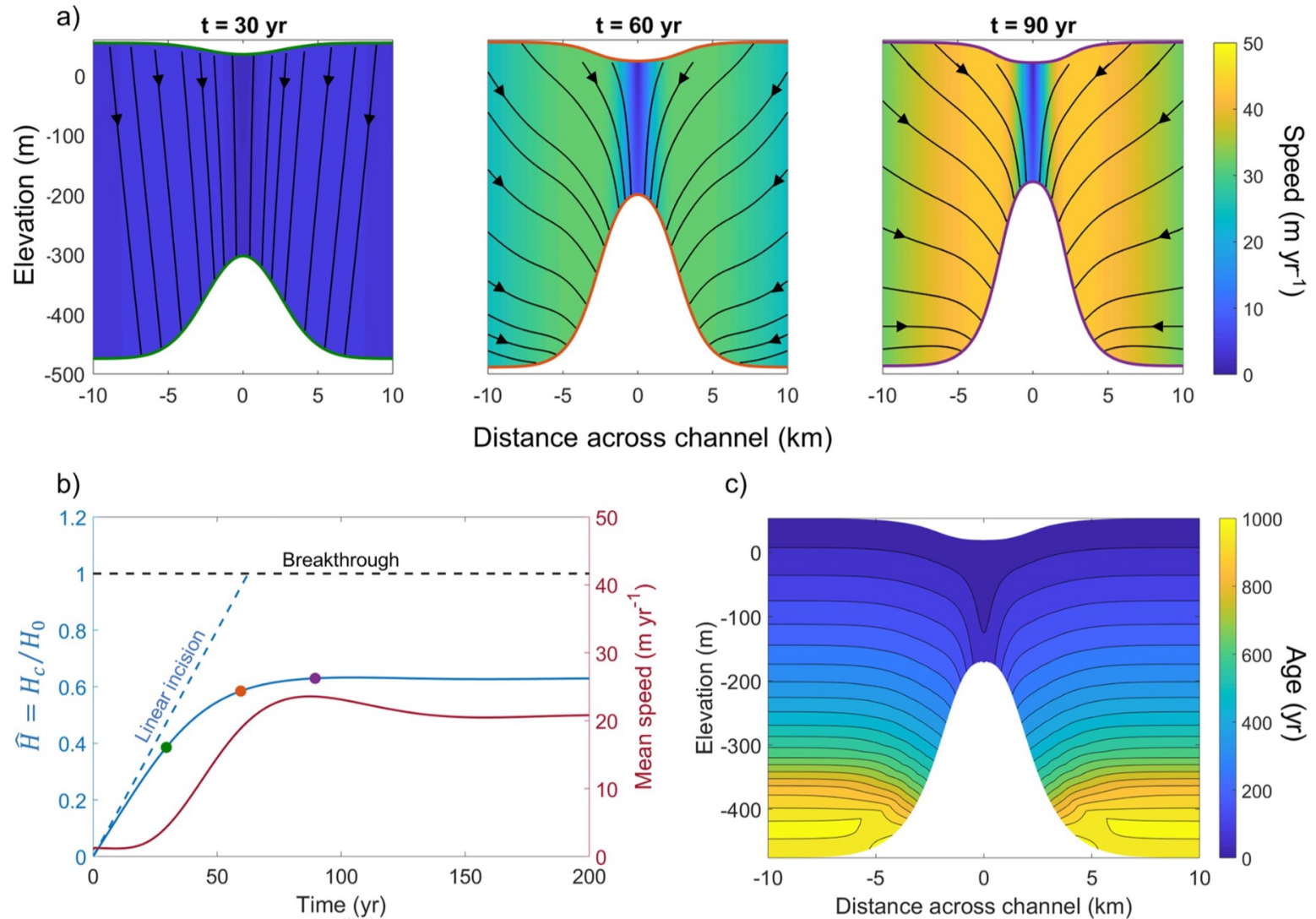
Plumes produce channels

Impacts of warm water on Antarctic ice shelf stability through basal channel formation

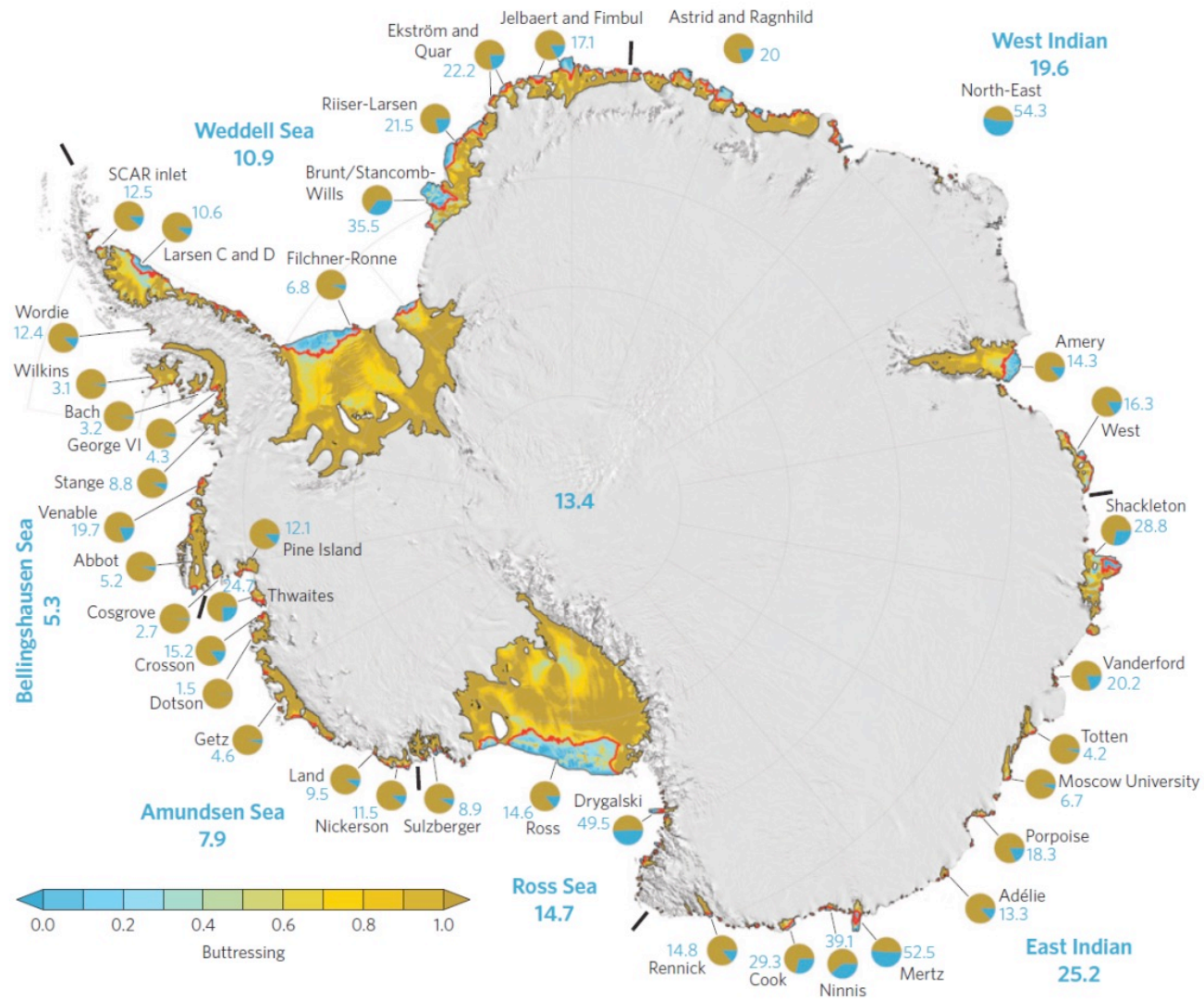
[Karen E. Alley](#) , [Ted A. Scambos](#), [Matthew R. Siegfried](#) & [Helen Amanda Fricker](#)



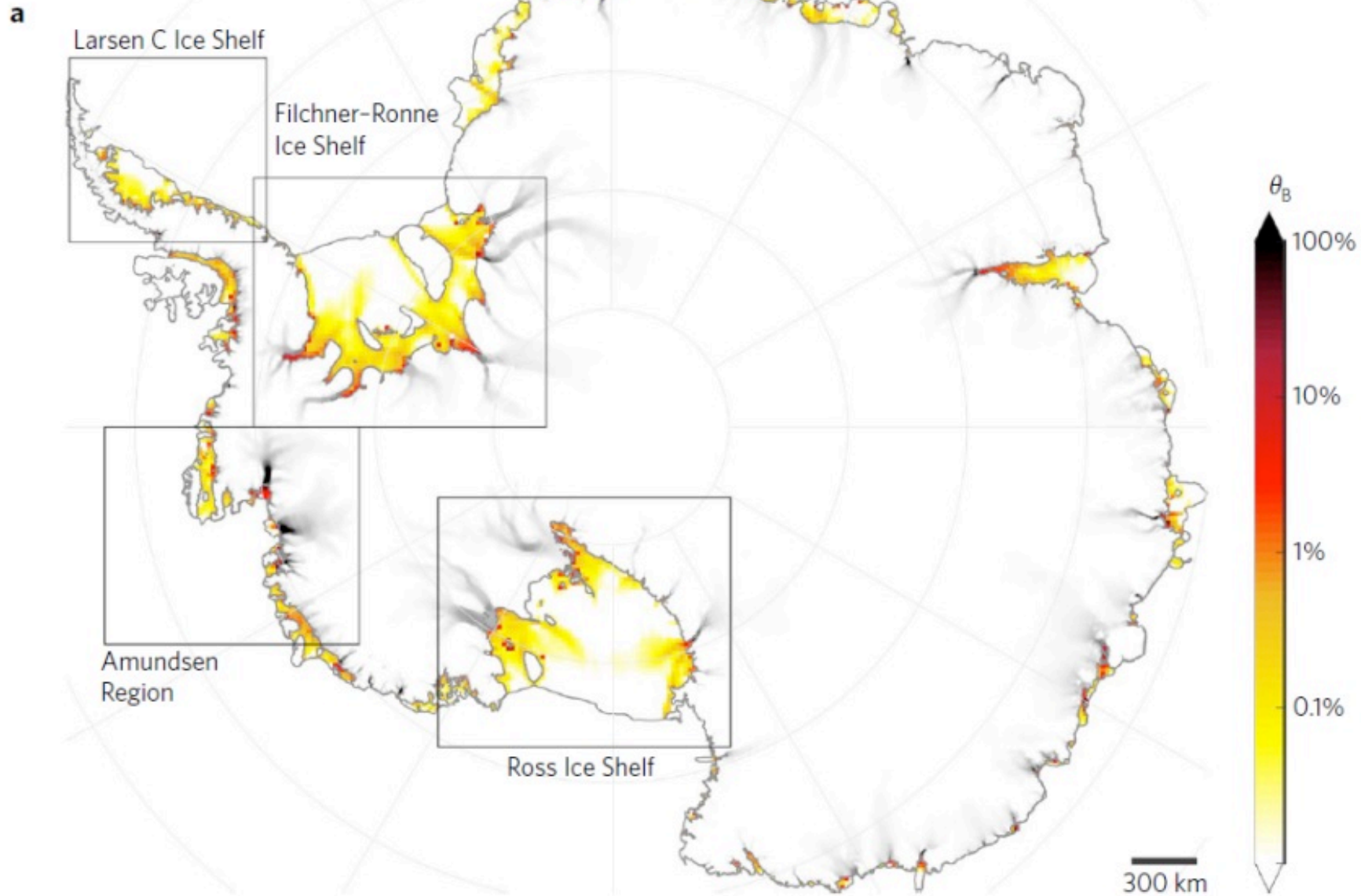
Plumes produce channels



Buttressing

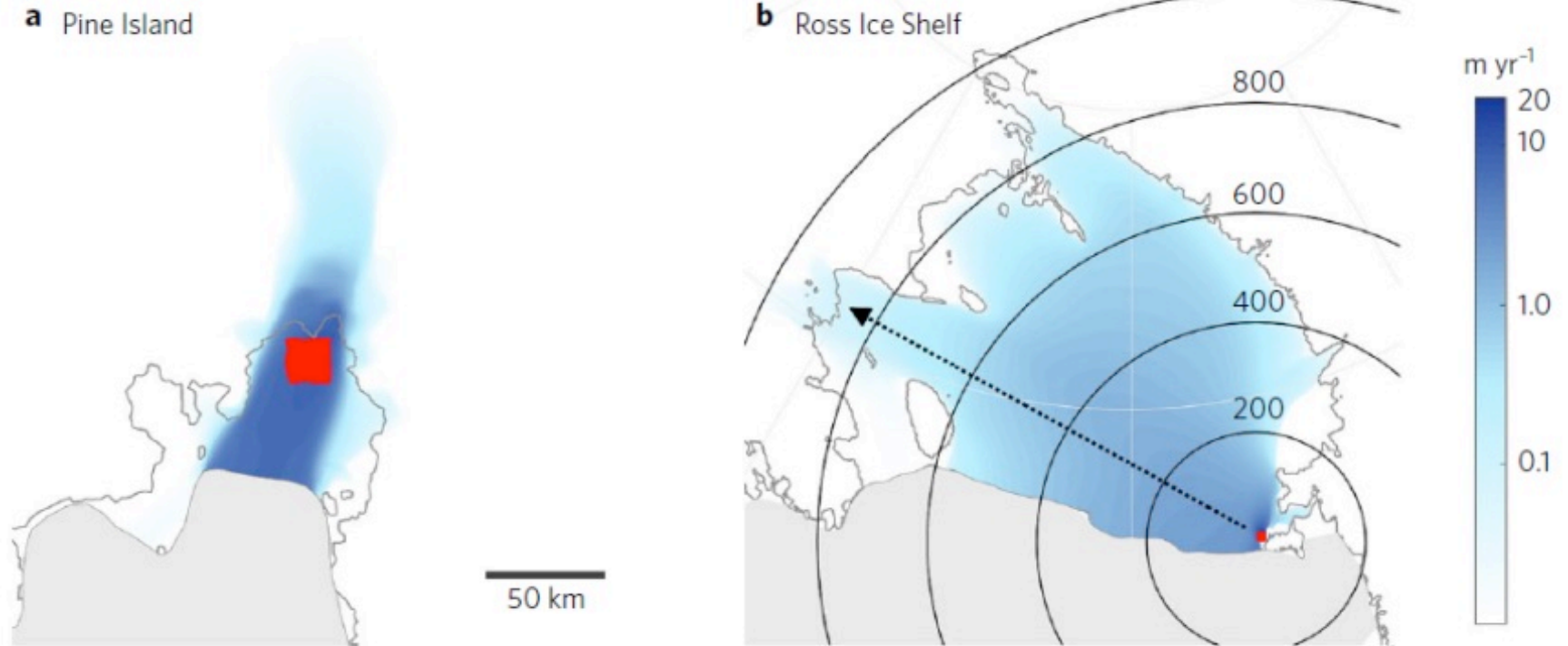


Buttressing



Reese et al. (2017)

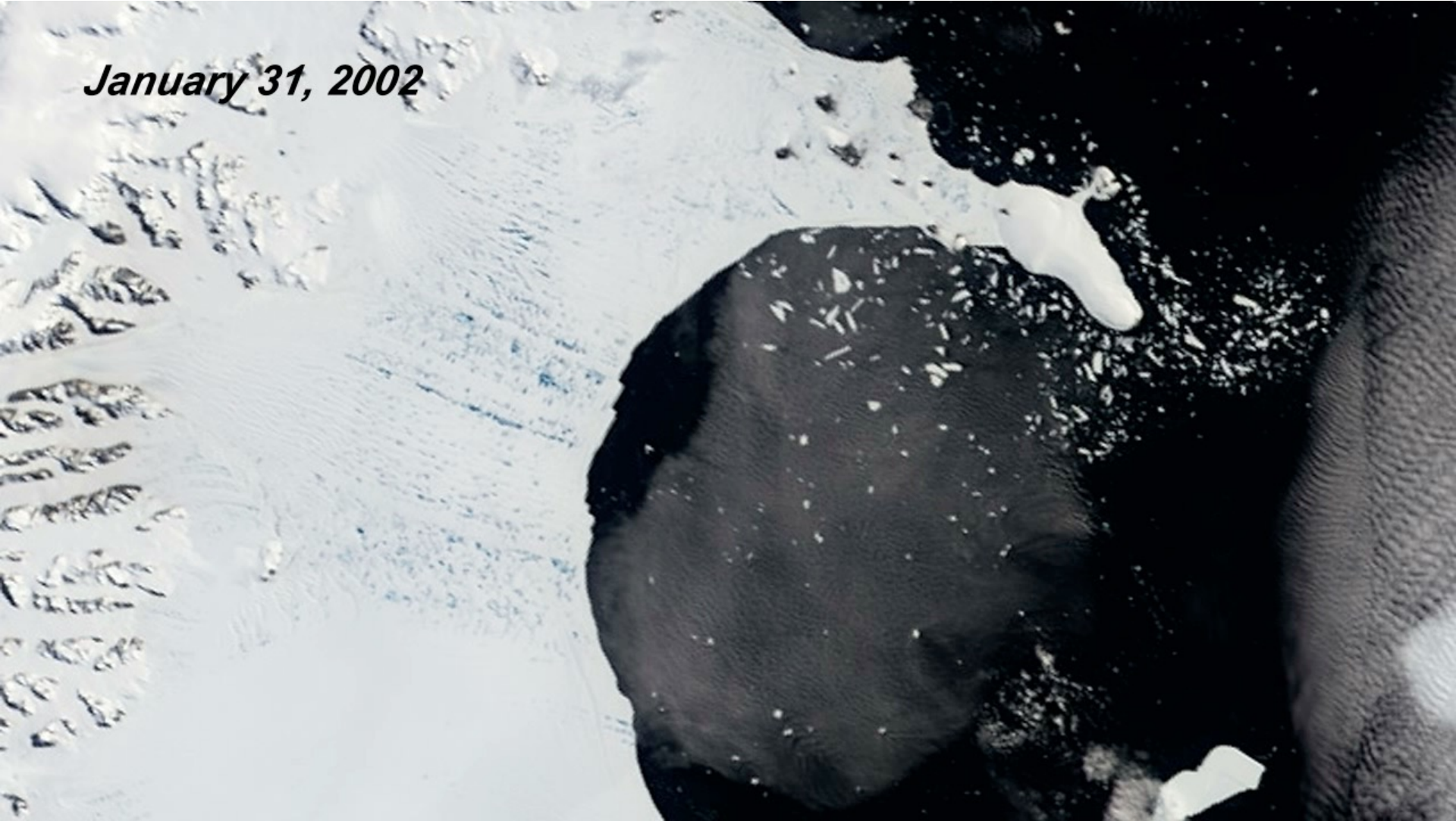
Buttressing



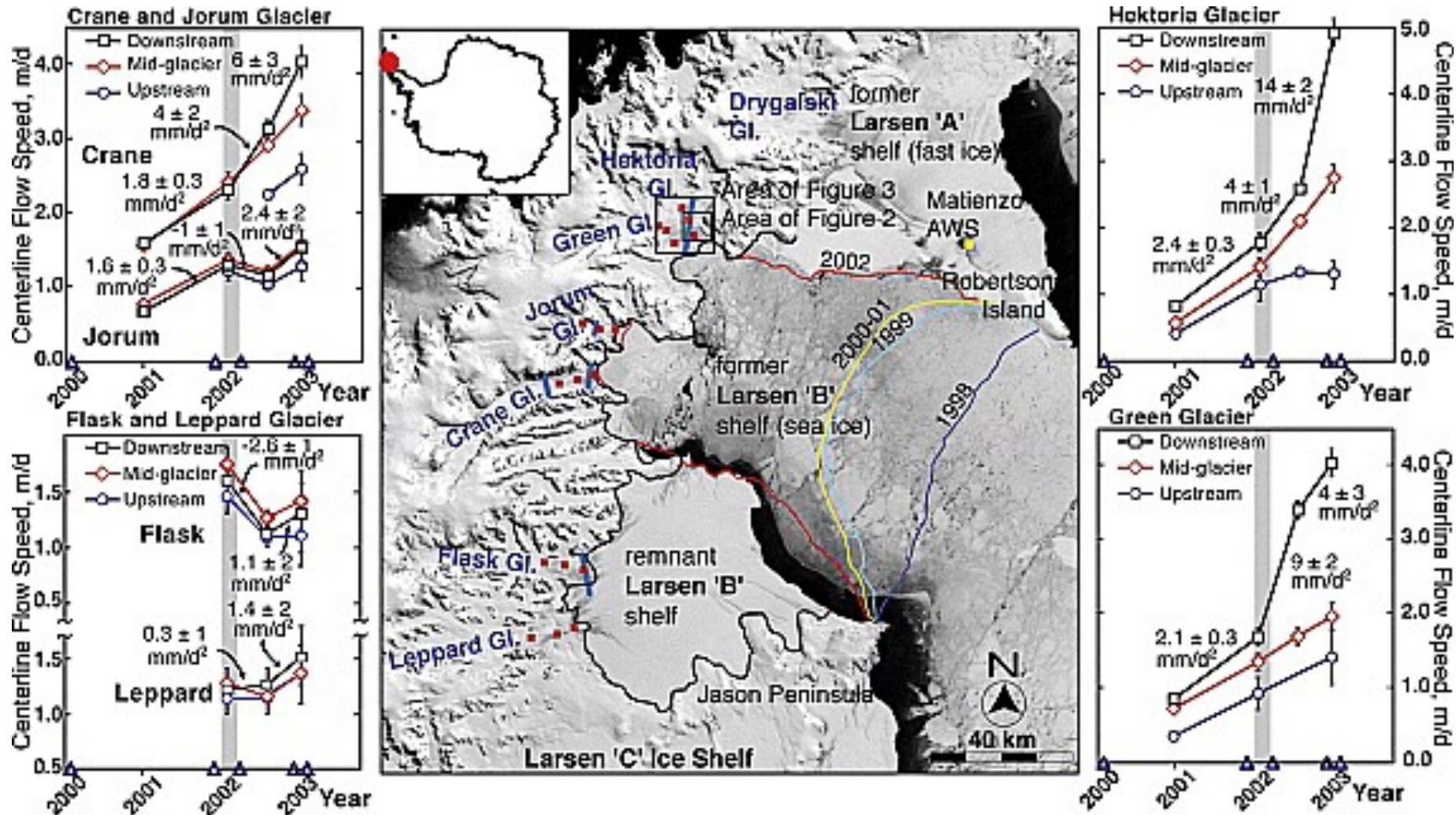
Buttressing

Larsen B Ice Shelf

January 31, 2002



Implications of buttressing loss

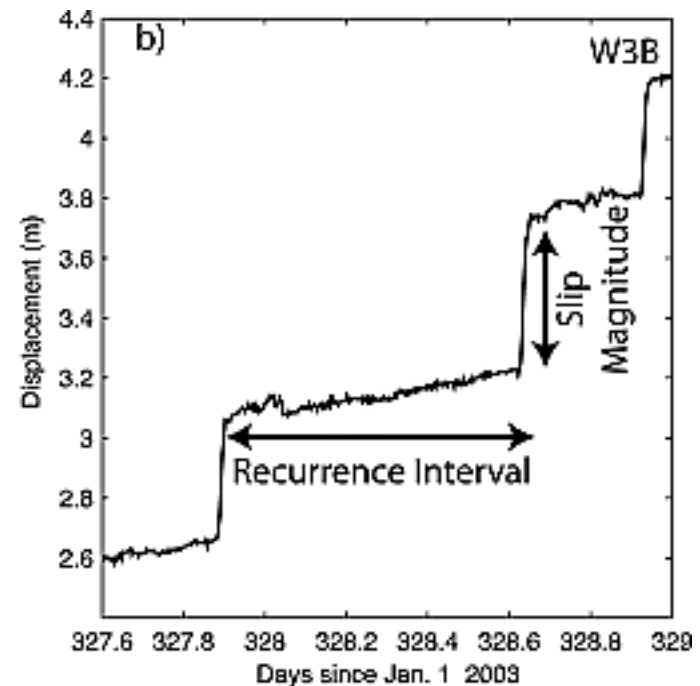
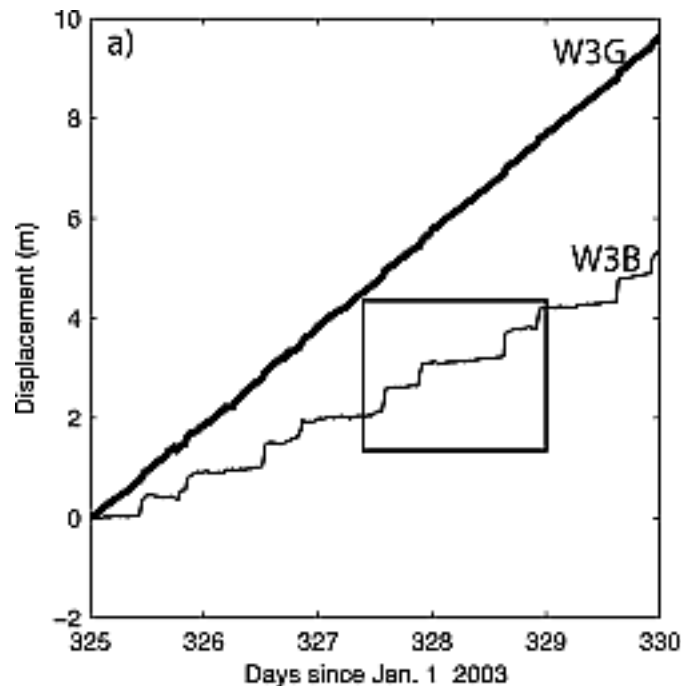


Scambos et al. (2004)

Tides - stick-slip motion

Basal mechanics of ice streams: Insights from the stick-slip motion of Whillans Ice Stream, West Antarctica

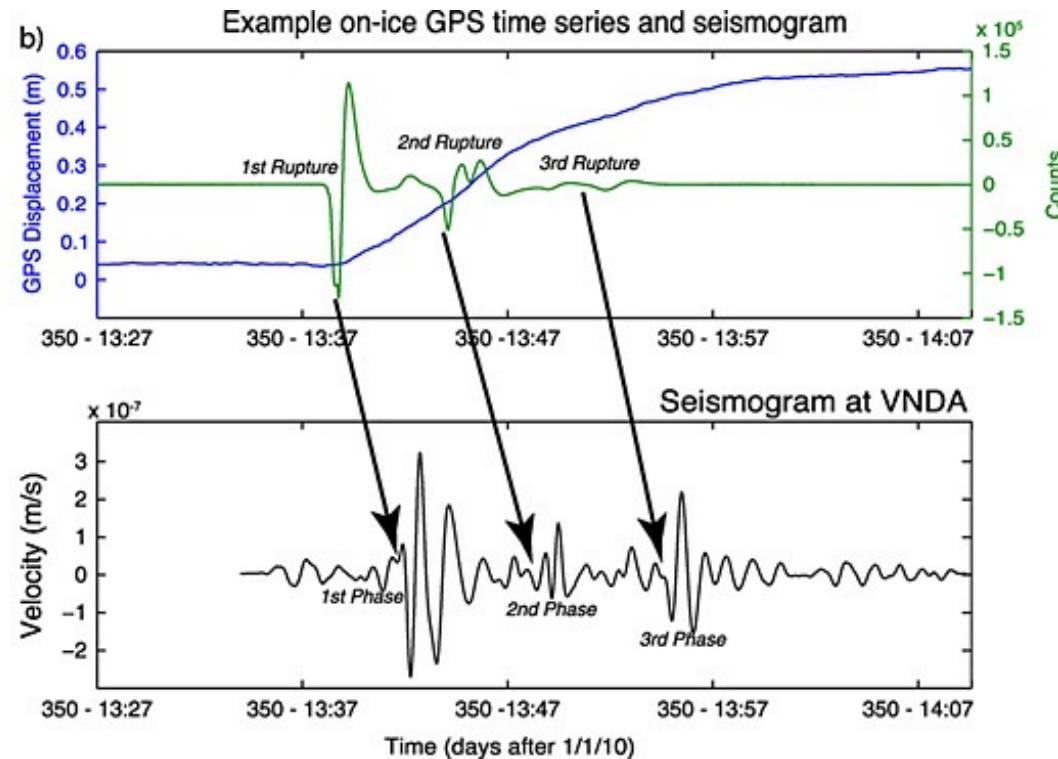
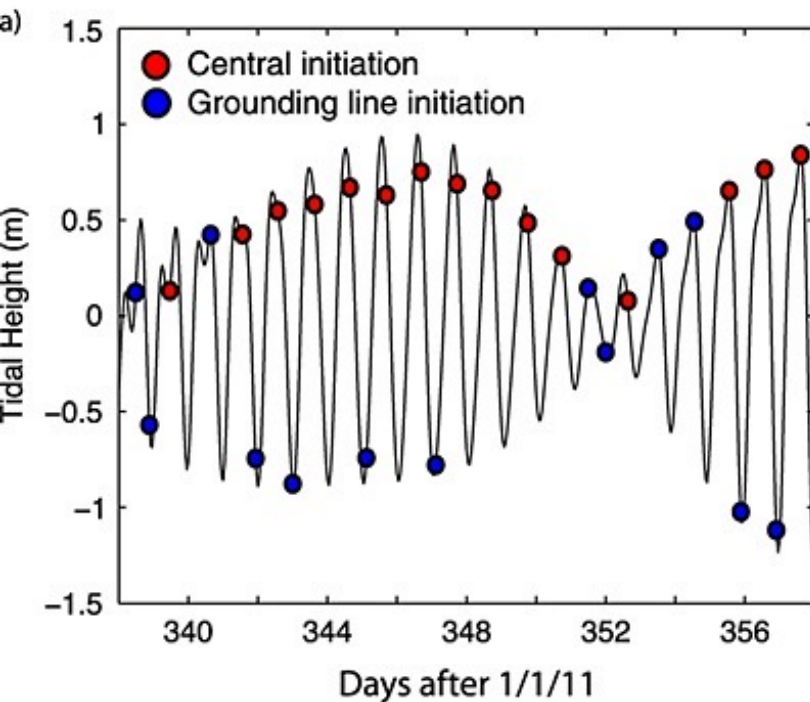
J. Paul Winberry ✉, Sridhar Anandakrishnan, Richard B. Alley, Robert A. Bindschadler, Matt A. King



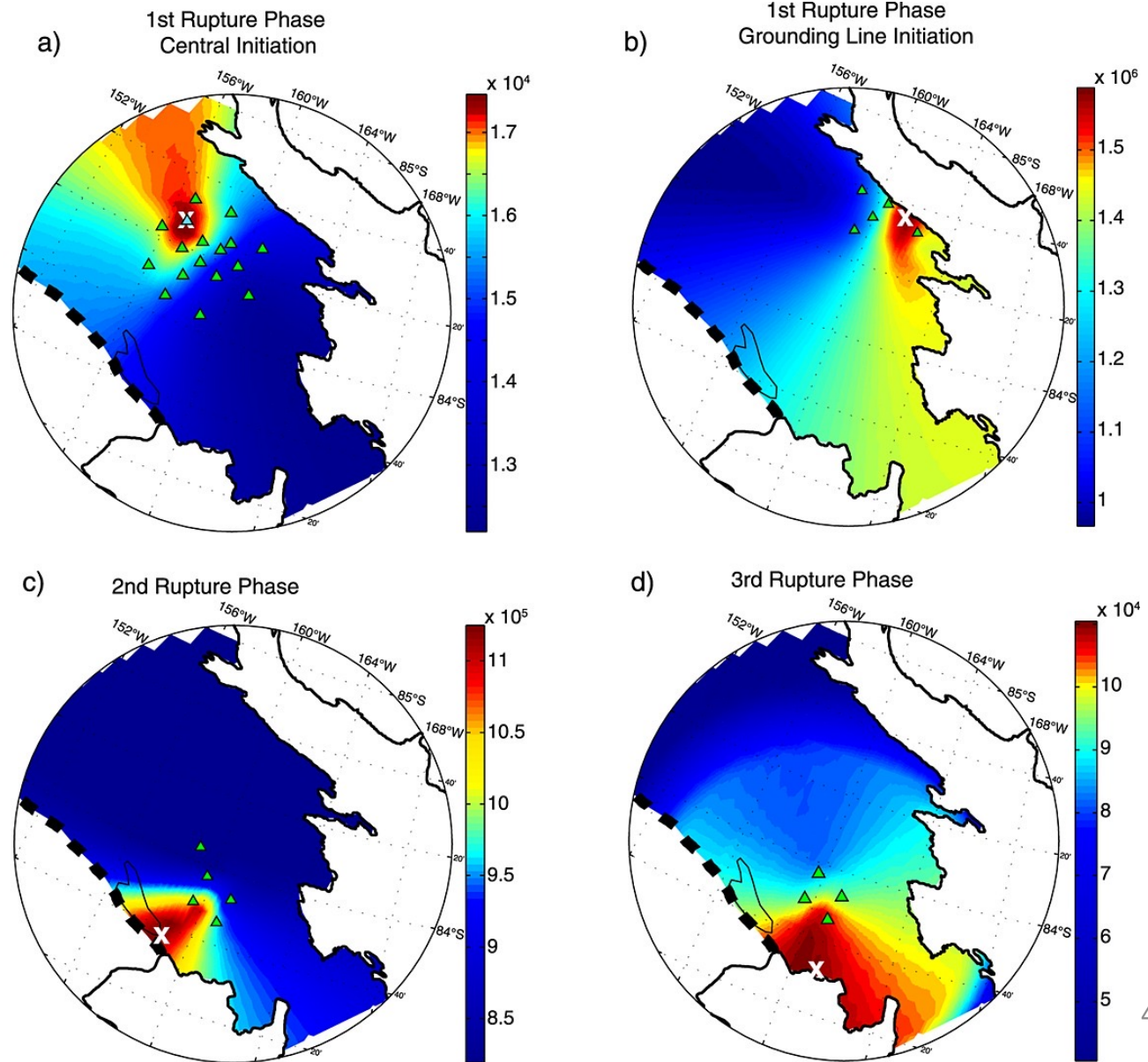
Tides - stick-slip motion

Seismic and geodetic evidence for grounding-line control of Whillans Ice Stream stick-slip events

Martin J. Pratt , J. Paul Winberry, Douglas A. Wiens, Sridhar Anandakrishnan, Richard B. Alley



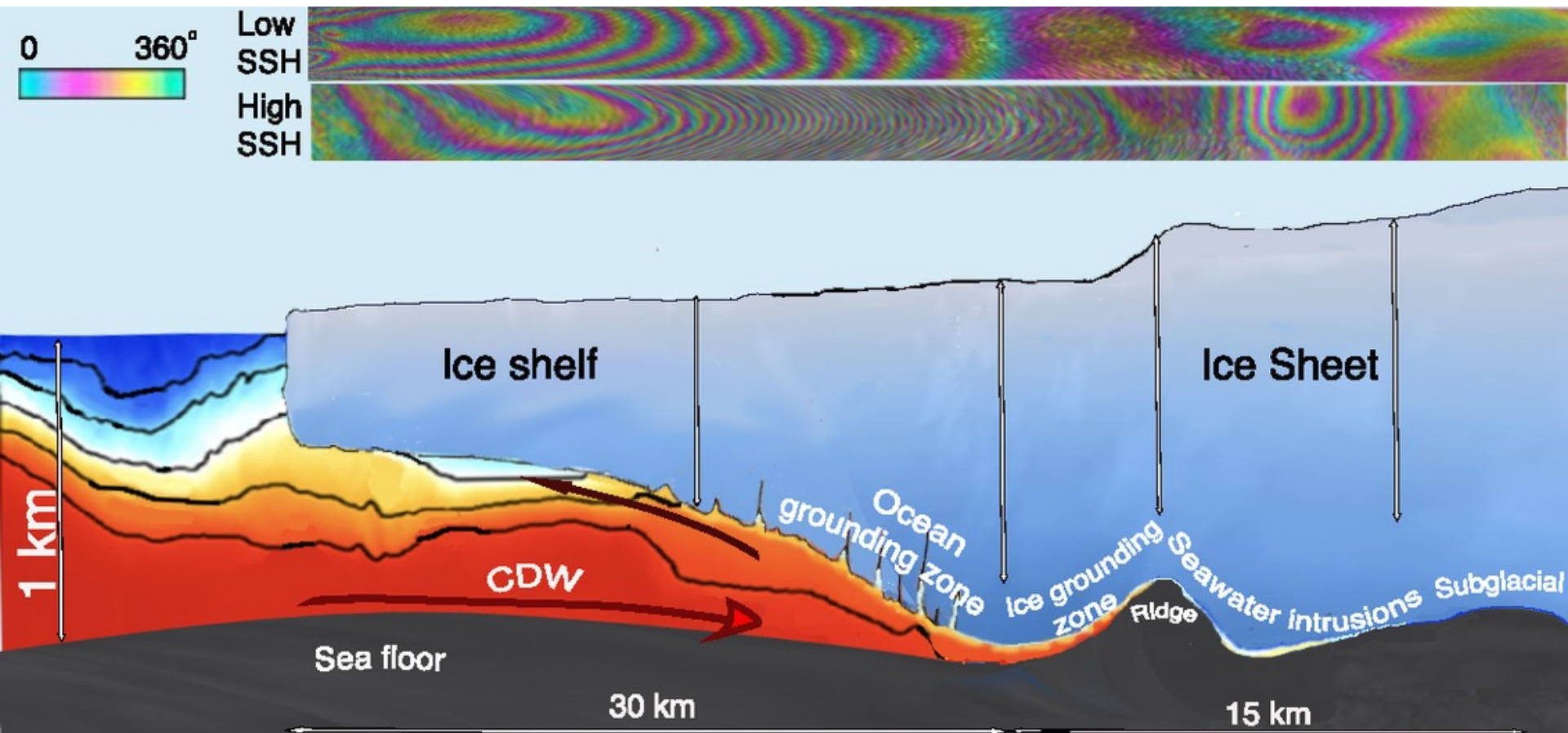
Tides - stick-slip motion



Seawater intrusion driven by tides

Widespread seawater intrusions beneath the grounded ice of Thwaites Glacier, West Antarctica

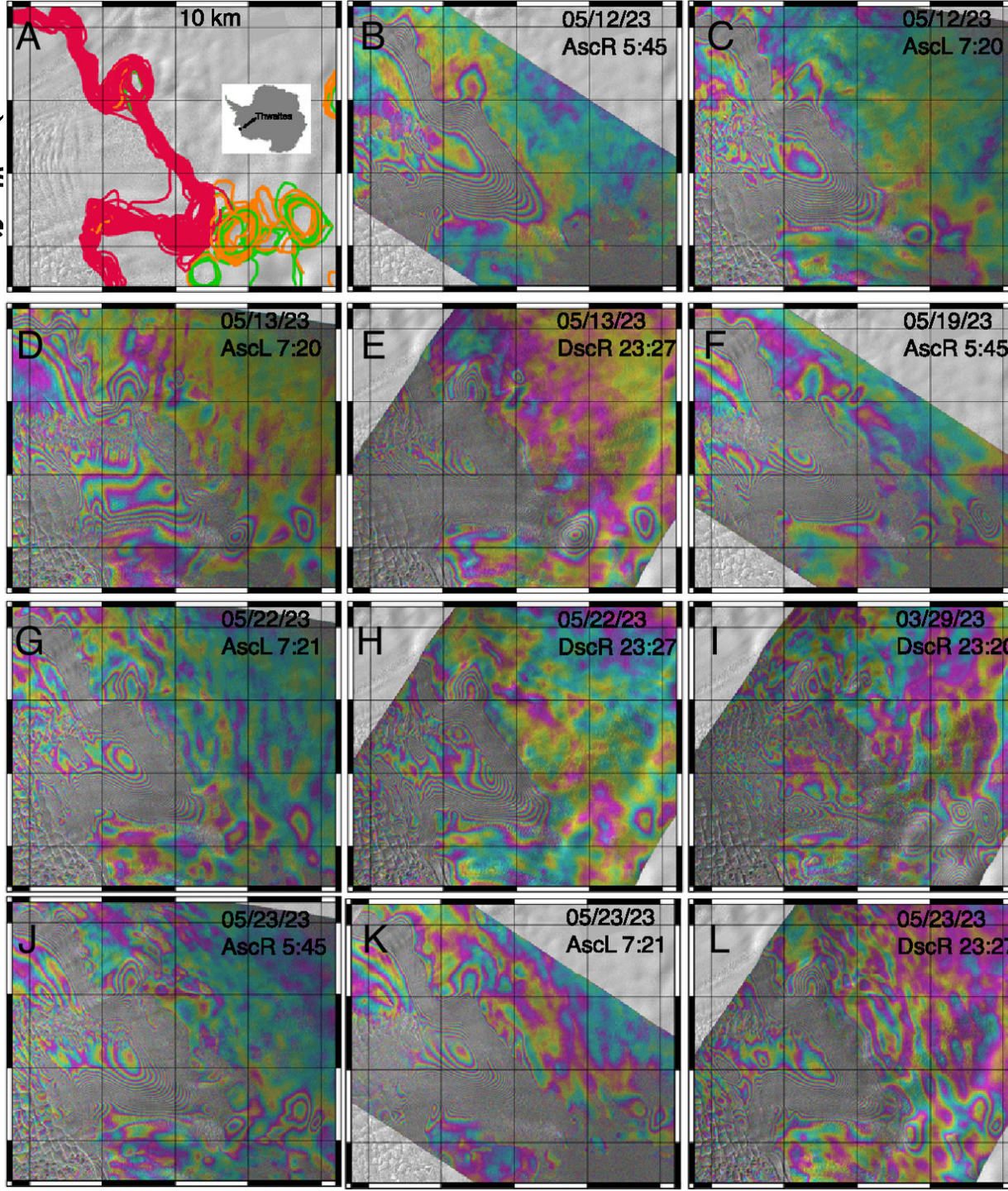
Eric Rignot , Enrico Ciraci , Bernd Scheuchl ,  +2, and Christine Dow [Authors Info & Affiliations](#)



Seawa

Widespread s
grounded ice
Antarctica

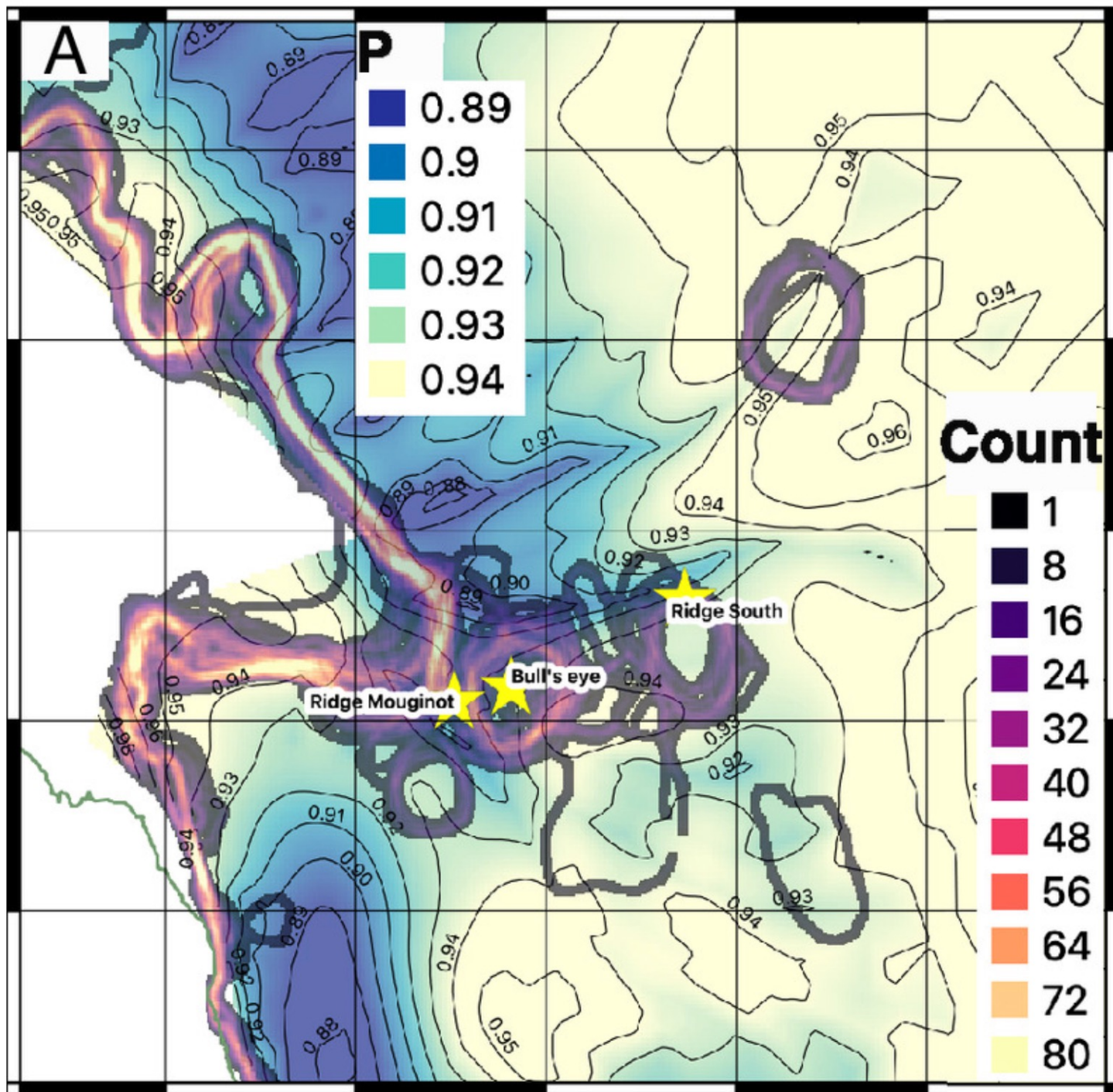
Eric Rignot  , Enrico Ciraci 



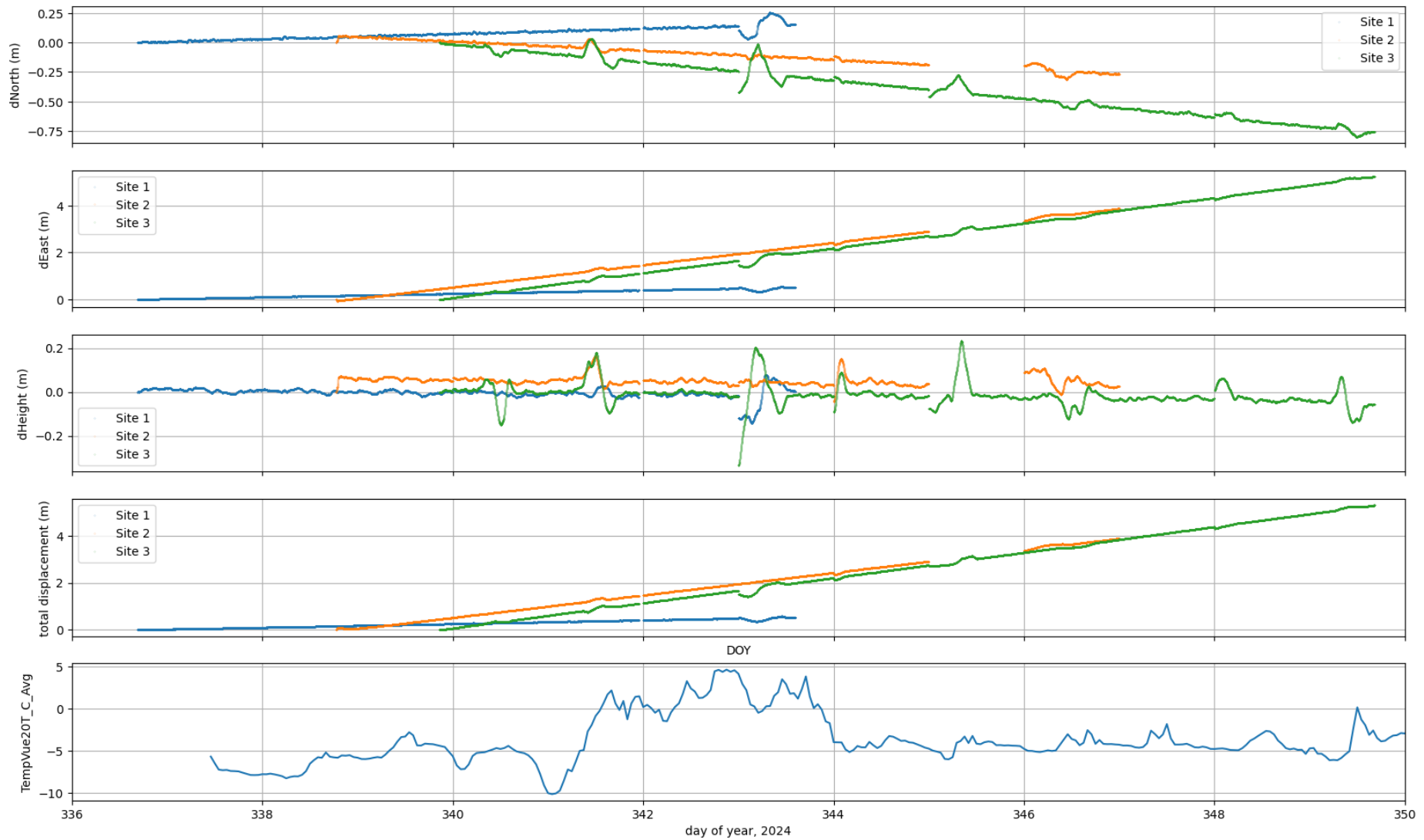
Seav

Widespr
grounde
Antarcti

Eric Rignot   , Em



Flask Glacier



References

- Alley, K.E., Scambos, T.A., Siegfried, M.R. and Fricker, H.A., 2016. Impacts of warm water on Antarctic ice shelf stability through basal channel formation. *Nature Geoscience*, 9(4), pp.290-293.
- Cook, S.J., Christoffersen, P., Todd, J., Slater, D. and Chauché, N., 2020. Coupled modelling of subglacial hydrology and calving-front melting at Store Glacier, West Greenland. *The Cryosphere*, 14(3), pp.905-924.
- Fürst, J.J., Durand, G., Gillet-Chaulet, F., Tavard, L., Rankl, M., Braun, M. and Gagliardini, O., 2016. The safety band of Antarctic ice shelves. *Nature Climate Change*, 6(5), pp.479-482.
- How, P., Schild, K.M., Benn, D.I., Noormets, R., Kirchner, N., Luckman, A., Vallot, D., Hulton, N.R. and Borstad, C., 2019. Calving controlled by melt-under-cutting: detailed calving styles revealed through time-lapse observations. *Annals of Glaciology*, 60(78), pp.20-31.
- Jenkins, A., 2011. Convection-driven melting near the grounding lines of ice shelves and tidewater glaciers. *Journal of Physical Oceanography*, 41(12), pp.2279-2294.
- Jenkins, A., Shoosmith, D., Dutrieux, P., Jacobs, S., Kim, T.W., Lee, S.H., Ha, H.K. and Stammerjohn, S., 2018. West Antarctic Ice Sheet retreat in the Amundsen Sea driven by decadal oceanic variability. *Nature Geoscience*, 11(10), pp.733-738.
- Reese, Ronja, G. Hilmar Gudmundsson, Anders Levermann, and Ricarda Winkelmann. "The far reach of ice-shelf thinning in Antarctica." *Nature Climate Change* 8, no. 1 (2018): 53.
- Rignot, E., Jacobs, S., Mouginot, J. and Scheuchl, B., 2013. Ice-shelf melting around Antarctica. *Science*, 341(6143), pp.266-270.
- Rignot, E., Ciraci, E., Scheuchl, B., Tolpekin, V., Wollersheim, M. and Dow, C., 2024. Widespread seawater intrusions beneath the grounded ice of Thwaites Glacier, West Antarctica. *Proceedings of the National Academy of Sciences*, 121(22), p.e2404766121.
- Pratt, M.J., Winberry, J.P., Wiens, D.A., Anandakrishnan, S. and Alley, R.B., 2014. Seismic and geodetic evidence for grounding-line control of Whillans Ice Stream stick-slip events. *Journal of Geophysical Research: Earth Surface*, 119(2), pp.333-348.
- Scambos, T., Fricker, H.A., Liu, C.C., Bohlander, J., Fastook, J., Sargent, A., Massom, R. and Wu, A.M., 2009. Ice shelf disintegration by plate bending and hydro-fracture: Satellite observations and model results of the 2008 Wilkins ice shelf break-ups. *Earth and Planetary Science Letters*, 280(1-4), pp.51-60.
- Slater, D.A., Nienow, P.W., Cowton, T.R., Goldberg, D.N. and Sole, A.J., 2015. Effect of near-terminus subglacial hydrology on tidewater glacier submarine melt rates. *Geophysical Research Letters*, 42(8), pp.2861-2868.
- Truffer, M. and Motyka, R.J., 2016. Where glaciers meet water: Subaqueous melt and its relevance to glaciers in various settings. *Reviews of Geophysics*, 54(1), pp.220-239.
- Wearing, M.G., Stevens, L.A., Dutrieux, P. and Kingslake, J., 2021. Ice-shelf basal melt channels stabilized by secondary flow. *Geophysical Research Letters*, 48(21), p.e2021GL094872.
- Winberry, J.P., Anandakrishnan, S., Alley, R.B., Bindshadler, R.A. and King, M.A., 2009. Basal mechanics of ice streams: Insights from the stick-slip motion of Whillans Ice Stream, West Antarctica. *Journal of Geophysical Research: Earth Surface*, 114(F1).